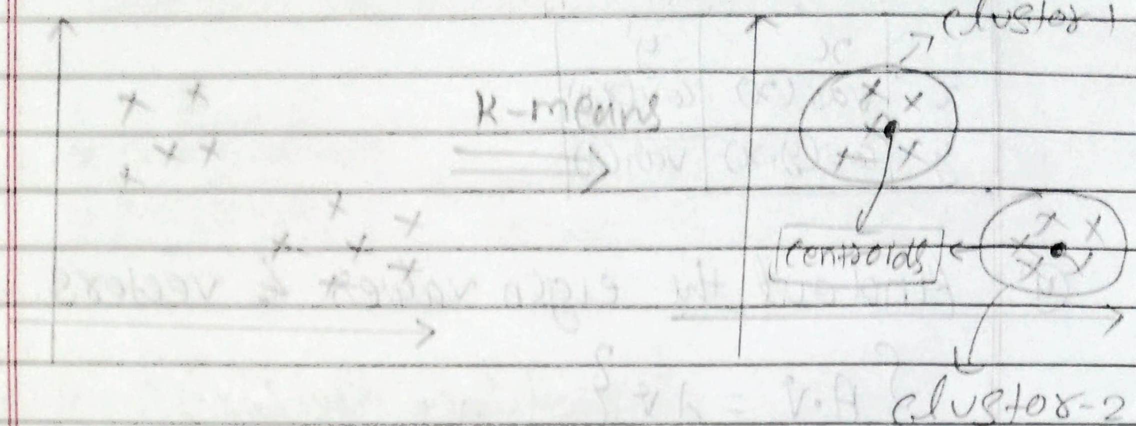
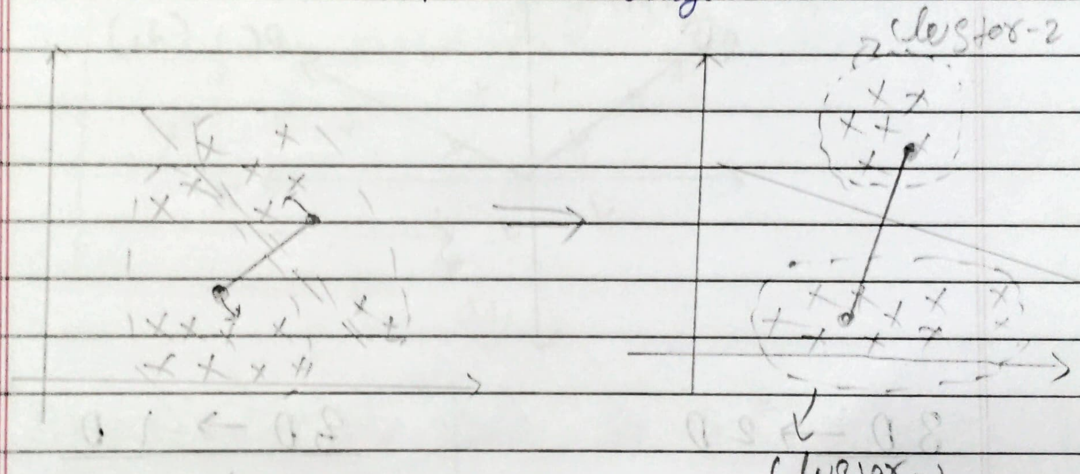
K- Means clustering Algorithm.



K-means

(i) Initialize some K - centroids
(ii) Points that are nearest to the centroids mark it in the same group.
(iii) move the centroid. → average

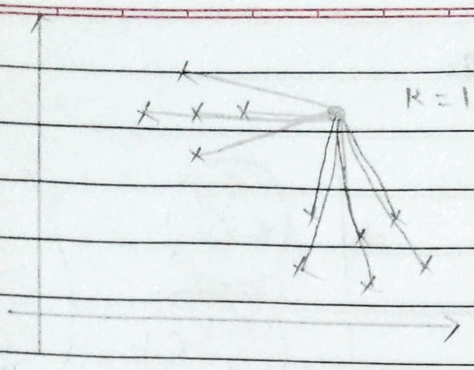
Repeat it again



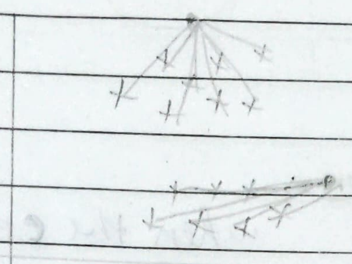
How do we select the K values?

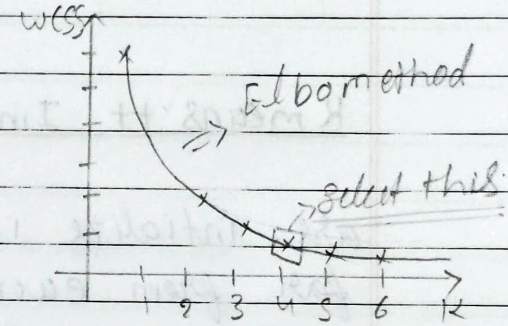
WCSS ⇒ within cluster sum of squares.

⇒ Initialize K = 1 to 20



$$WCSS = \sum_{i=1}^{n} \left(\text{distance b/w point to nearest centroid} \right)^2$$





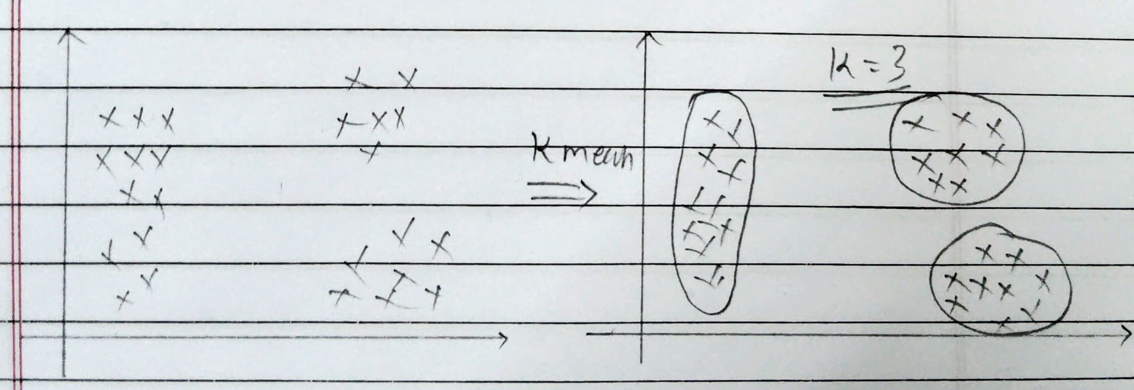
⇒ Eucledian distance

$$\left\{ D = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2} \right\}$$

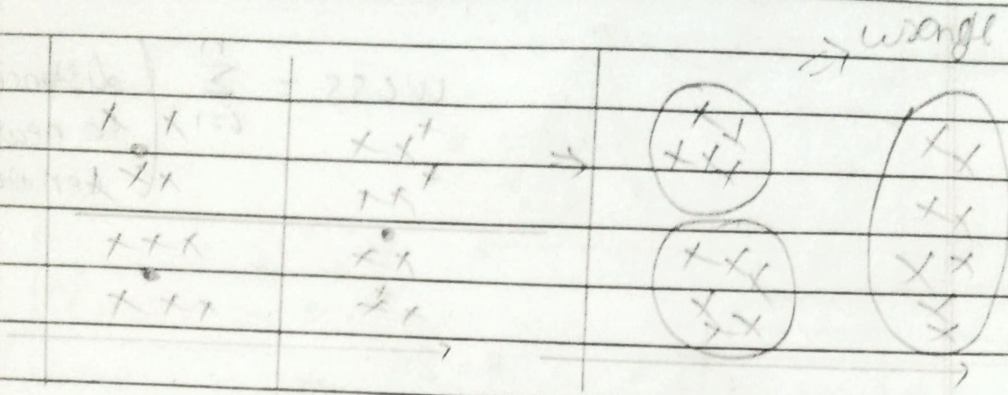
⇒ manhattan distance

$$\left\{ D = |x_2 - x_1| + |y_2 - y_1| \right\}$$

⇒ Random Initialization Trap (K means ++)



Random Initialization



K means ++ Initialization =>

when the centrodieds are intialize in K means ++ it is initialize far from each other for the better and accurate initialization.